

Chapter 6: AI Applications

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Syllabus

6.1. Expert Systems: Characteristics, Architecture, Development and Various Applications

6.2 NLP: Level of Analysis and Application

6.3. Robotics & Computer Vision: Fundamental, Components and Applications

6.4. Sustainable AI systems

Hrs: 7 hrs

Expert Systems

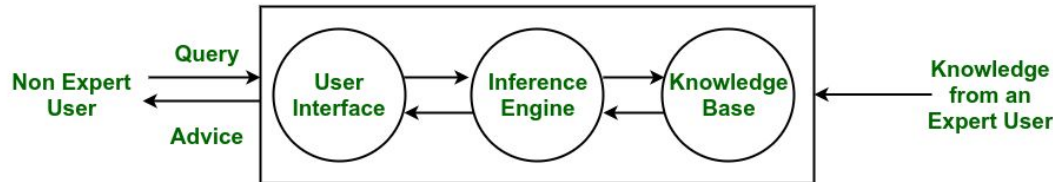
- ❖ Expert systems are a crucial subset of artificial intelligence (AI) that simulate the decision-making ability of a human expert.
- ❖ These systems use a knowledge base filled with domain-specific information and rules to interpret and solve complex problems.
- ❖ The key idea behind expert systems is to preserve and replicate human expertise.
- ❖ **Importance:**
 - **Preserving Expertise:** They capture the knowledge of human experts and store it in a digital format. This ensures that valuable expertise isn't lost when an expert retires or leaves.
 - **Improving Decision-Making:** By relying on data and rules, expert systems provide consistent and unbiased recommendations.
 - **Saving Time and Money:** They automate tasks that would otherwise require human intervention, reducing costs and increasing efficiency.
 - **Accessibility:** Expert systems make expert-level knowledge available to non-experts, democratizing access to specialized information.

Expert System: Architecture

- ❖ An expert system is made up of several interconnected components
 - **Knowledge Base:** The Heart of the System
 - The knowledge base is the heart of an expert system.
 - It contains all the facts, rules, and expert knowledge related to a specific domain.
 - Think of it as a library filled with textbooks, research papers, and expert opinions.
 - **Inference Engine:** The Brain Behind the Decisions
 - The inference engine is the brain of the expert system.
 - It processes the information stored in the knowledge base to draw conclusions or make recommendations.
 - The inference engine uses reasoning strategies (like forward chaining or backward chaining) to analyze data and apply rules.
 - **User Interface:** Bridging the Gap Between System and User
 - The user interface is the bridge that allows users to interact with the expert system.
 - It's designed to be intuitive and user-friendly, ensuring that even non-experts can use the system effectively.
 - Users provide a query (problem or question), and the system processes the request.

Expert System: Architecture

- ❖ An expert system is made up of several interconnected components
 - **Explanation Module:** Building Trust Through Transparency
 - The explanation module is a critical feature that explains how the system arrived at a particular conclusion.
 - It's like a teacher showing their work when solving a math problem.
 - This module provides users with a clear, step-by-step explanation of the system's reasoning.
 - **Knowledge Acquisition Module:** Keeping the System Up-to-Date
 - The knowledge acquisition module is responsible for updating and expanding the knowledge base.
 - It ensures that the system stays current with the latest information and trends.
 - Without regular updates, the system's knowledge base can become outdated, reducing its effectiveness.



Export System: Pros and Cons

Benefits

- ❖ Consistency:
- ❖ Availability:
- ❖ Cost-Effectiveness
- ❖ Knowledge Preservation:

Limitations

- ❖ Knowledge Limitation
- ❖ Lack of Flexibility
- ❖ Maintenance

Expert System: Development Phase

The five development stages/phase of the expert system

❖ Problem Identification

- The development process begins with Problem Identification, where the **primary goal is to determine if an Expert System is actually the right tool for the job.**
- In this stage, the project team—usually consisting of stakeholders and knowledge engineers—analyzes the problem to ensure it is narrow enough to be manageable but complex enough to require genuine human expertise.

❖ Conceptualization

- Once the problem is defined, the team moves into Conceptualization, which is arguably the most human-centric phase of the cycle.
- During this stage, the **knowledge engineer interacts** intensively with the **domain expert** to **uncover the strategies, rules** of thumb, and **specific constraints** used to solve problems.
- The objective here is to create a diagram or model of the expert's thought process without worrying about computer code yet.

Expert System: Development Phase

❖ Formalization

- After mapping the expert's thoughts, the Formalization phase involves translating these abstract concepts into a structured format that a computer can logically process.
- The knowledge engineer selects a specific knowledge representation method, such as "If-Then" rules, frames, or semantic networks, to organize the data gathered in the previous step.

❖ Implementation

- The Implementation phase is where the actual construction of the system takes place, turning the formal specifications into a working software prototype.
- The knowledge engineer codes the Knowledge Base (the repository of facts and rules) and configures the Inference Engine (the processing unit that applies the rules to the data).
- They also build the user interface that will allow non-experts to interact with the system.

❖ Testing and Validation

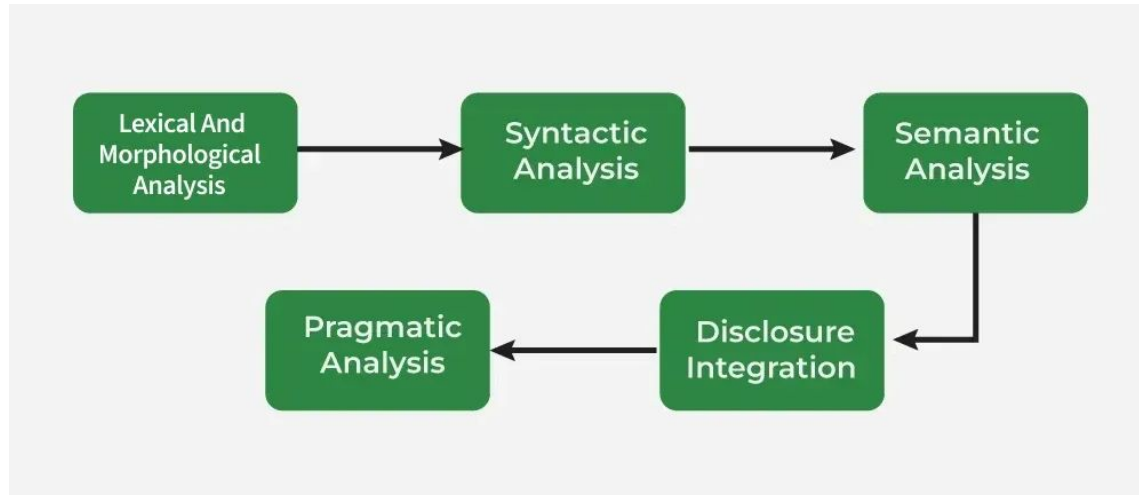
- Once the prototype is built, it undergoes rigorous Testing and Validation to ensure it performs at the level of a human expert.
- In this stage, the system is fed real-world case studies, and its answers are compared against the solutions provided by the domain expert. This is a critical feedback loop; if the system makes a mistake, the knowledge engineer must trace the error back to a specific rule or logic gap and correct it.

Expert System: Applications

- ❖ **Medical Diagnosis:** Expert systems assist doctors by analyzing symptoms and medical history to suggest possible diagnoses or treatment options. For example, MYCIN, an early expert system, helped identify bacterial infections and recommend antibiotics.
- ❖ **Financial Services:** In finance, expert systems are used for credit scoring, fraud detection, and investment advice. They analyze financial data and patterns to make informed decisions.
- ❖ **Technical Support:** Expert systems can troubleshoot and provide solutions for technical issues. They guide users through problem-solving steps based on pre-defined rules and knowledge.
- ❖ **Manufacturing:** In manufacturing, expert systems help optimize production processes, perform quality control, and manage inventory by analyzing data and making recommendations.

Natural Language Processing (NLP)

- ❖ Natural Language Processing (NLP) helps computers to understand, analyze and interact with human language.
- ❖ It involves a series of phases that work together to process language and each phase helps in understanding structure and meaning of human language.



NLP: Levels of Analysis

❖ Lexical and Morphological Analysis

➤ Lexical Analysis

- It focuses on identifying and processing words (or lexemes) in a text.
- It breaks down the input text into individual tokens that are meaningful units of language such as words or phrases.
- Tasks:
 - Tokenizations: example: ["I", "am", "reading", "a", "book"]
 - Part of Speech: example ["I" → Pronoun (PRP), "am" → Verb (VBP), "reading" → Verb (VBG), "a" → Article (DT), "book" → Noun (NN)]

➤ Morphological Analysis

- It deals with morphemes which are the smallest units of meaning in a word.
- It is important for understanding structure of words and their parts by identifying free morphemes (independent words like "cat") and bound morphemes (like prefixes or suffixes e.g. "un-" or "-ing").
- Task: Stemming and Lemmatization:
- It helps in breaking the composition of complex words.

NLP: Levels of Analysis

❖ Syntactic Analysis (Parsing)

- Syntactic Analysis helps in understanding how words in a sentence are arranged according to grammar rules.
- It ensures that the sentence follows correct grammar which makes the meaning clearer.
- The goal is to create a parse tree which is a diagram showing the structure of sentence.
- Key components:
 - POS Tagging: Assigning parts of speech (noun, verb, adjective) to words in a sentence as discussed earlier.
 - Ambiguity Resolution: Handling words that have multiple meanings (e.g "book" can be a noun or a verb)
- Examples:
 - Correct Syntax: "John eats an apple."
 - Incorrect Syntax: "Apple eats John an."

NLP: Levels of Analysis

❖ Semantic Analysis

- Semantic Analysis focuses on understanding meaning behind words and sentences.
- It ensures that the text is not only grammatically correct but also logically coherent and contextually relevant.
- It aims to understand dictionary definitions of words and their usage in context and also find whether the arrangement of words in a sentence makes logical sense.
- Key Tasks in Semantic Analysis
 - **Named Entity Recognition (NER):** It identifies and classifies entities such as names of people, locations, organizations, dates and more. These entities provide important meaning in the text and help in understanding the context.
 - **Word Sense Disambiguation (WSD):** Many words have multiple meanings depending on the context in which they are used. It identifies the correct meaning of a word based on its surrounding text.
- Example: "Apple eats a John." while grammatically correct this sentence doesn't make sense semantically because an apple cannot "eat" a person. Semantic analysis ensures that the meaning is logically sound and contextually appropriate

NLP: Levels of Analysis

❖ Discourse Integration

- It is the process of understanding how individual sentences or segments of text connect and relate to each other within a broader context.
- This phase ensures that the meaning of a text is consistent and coherent across multiple sentences or paragraphs.
- Key aspects of discourse integration:
 - Anaphora Resolution: Anaphora refers to the use of pronouns or other references that depend on earlier parts of the text. For example in the sentence "Taylor went to the store. She bought groceries" pronoun "She" refers back to "Taylor." It ensures that references like these are correctly understood by linking them to their antecedents.
 - Contextual References: Many words or phrases can only be fully understood when considered in the context of following sentences. It helps in interpreting how certain words or phrases focuses on context. For example "It was a great day" is clearer when you know what event or situation is being discussed.
- Example of Discourse Integration
 - "Taylor went to the store to buy some groceries. She realized she forgot her wallet."
Understanding that "Taylor" is the antecedent of "she" is important for understanding sentence's meaning.

NLP: Levels of Analysis

❖ Pragmatic Analysis

- Pragmatic analysis helps in understanding the deeper meaning behind words and sentences by looking beyond their literal meanings.
- While semantic analysis looks at the direct meaning it considers the speaker's or writer's intentions, tone and context of the communication.
- Key tasks in pragmatic analysis:
 - Understanding Intentions: Sometimes language doesn't mean what it says literally. For example when someone asks "Can you pass the salt?" it's not about ability but a polite request. It helps to understand true intention behind such expressions.
 - Figurative Meaning: Language often uses idioms or metaphors that can't be taken literally.
- Examples of Pragmatic Analysis
 - "Hello! What time is it?" here it might be a straightforward request for the current time but it could also imply concern about being late.
 - For example "I'm falling for you" means "I love you" not literally falling. It helps to interpret these non-literal meanings.

NLP: Challenges

❖ Ambiguity

- Ambiguity is the biggest hurdle in NLP, occurring when a sentence can be interpreted in multiple ways. This happens at the lexical level (one word having multiple meanings) or the structural level.
- Example: "I saw the man with the telescope." Does this mean I used a telescope to see the man, or did I see a man who was holding a telescope? The machine struggles to decide without more context.

❖ Sarcasm and Irony

- Machines are generally literal and struggle to detect when the intended meaning is the opposite of the literal meaning. Sarcasm relies heavily on tone (in speech) or world knowledge (in text), which AI often lacks.
- Example: "Oh, great! Another flat tire." A sentiment analysis model might see the word "great" and classify this as a positive sentence, completely missing the sarcasm.

❖ Slang and Neologisms

- Language changes faster than dictionaries do. New slang, internet abbreviations, and cultural references appear constantly.
- Example: If a user tweets, "That movie was straight fire," an older NLP model might interpret "fire" as a combustion hazard rather than the slang meaning of "excellent."

❖ Domain-Specific Language

- A model trained on Wikipedia articles might fail when processing medical records or legal contracts because the vocabulary and sentence structures are vastly different.
- Example: The word "bug" means something very different to a software engineer (an error) than to a biologist (an insect).

Embeddings & Modern NLP Models

❖ Word Embeddings (Word2Vec, GloVe)

- Traditional NLP treated words as simple ID numbers. Embeddings revolutionized this by representing words as "vectors" (lists of numbers) in a multi-dimensional space. The key idea is that words with similar meanings are placed closer together in this mathematical space.
 - Word2Vec: Uses neural networks to learn word associations from a large corpus of text. It can capture relationships like "King - Man + Woman = Queen."
 - GloVe (Global Vectors): Focuses on how frequently words appear together in the entire dataset to build these relationships.

❖ Contextual Embeddings (BERT, ELMo)

- The older embeddings (like Word2Vec) had a flaw: the word "bank" had the same vector whether it meant a river bank or a money bank. Modern models use contextual embeddings.
- BERT (Bidirectional Encoder Representations from Transformers): This model reads text in both directions (left-to-right and right-to-left) at once. This allows it to understand that in "bank account," bank refers to finance, but in "river bank," it refers to land.

Modern NLP Models

- ❖ The Transformer Architecture
- ❖ Generative Pre-trained Transformers
 - GPT Series
 - LLaMA
 - Claude
 - Deepseek
 - Proximity Lab
- ❖ Multimodal Models
 - Gemini
 - GPT-4o:

Robotics and its Fundamental Components

- ❖ Robotics is the interdisciplinary branch of engineering and science that includes mechanical engineering, electrical engineering, computer science, and others.
- ❖ It deals with the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing.
- ❖ A robot is essentially a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks.
- ❖ The fundamental components that make up a robot are distinct but interconnected:
 - Sensors: These are the "eyes and ears" of the robot. They gather information about the internal state of the robot or the outside environment. For example, a Lidar sensor on a self-driving car measures the distance to obstacles, while a proprioceptive sensor in a robotic arm tells the system the exact angle of its joints.

Robotics and its Fundamental Components

- ❖ **Actuators:** These are the "muscles" responsible for moving the robot. They convert stored energy into movement. Common examples include DC motors for rotating wheels, servo motors for precise joint angling in a humanoid robot, or hydraulic cylinders for heavy lifting in industrial excavators.
- ❖ **Control System:** This is the "brain" that processes sensor data and tells the actuators what to do. It can range from a simple microcontroller on a toy robot to a complex AI supercomputer on a Mars rover that plans paths autonomously.
- ❖ **Power Supply:** The energy source, which could be batteries (for mobile robots like drones), tethered electrical cables (for factory robots), or even solar panels (for space rovers).
- ❖ **End Effectors:** These are the "hands" or tools attached to the robot's arm to interact with the environment. Examples include a gripper for picking up boxes, a welding torch for car assembly, or a suction cup for handling glass.

Robot Classifications

Robots are classified based on their mechanical configuration and their intended application. The most common classification is by kinematics (how they move):

- ❖ **Articulated Robots:** These look like a human arm and are the most common in manufacturing. They typically have 6 rotary joints (6 degrees of freedom), allowing them to reach objects from any angle.
- ❖ **SCARA Robots:** Standing for "Selective Compliance Assembly Robot Arm," these are rigid in the vertical axis but flexible horizontally. They are perfect for "pick-and-place" tasks where an object needs to be moved from a conveyor belt to a box quickly.
- ❖ **Cartesian (Gantry) Robots:** These move in straight lines along X, Y, and Z axes. They are often used for 3D printing or CNC machining because they are very precise and stable.
- ❖ **Mobile Robots:** These are not fixed to one location. This category includes AGVs (Automated Guided Vehicles) in warehouses that follow magnetic strips on the floor, and AMRs (Autonomous Mobile Robots) that navigate freely using maps

Computer Graphics vs. Image Processing vs. Computer Vision

- ❖ **Computer Graphics:** This is the process of creating images from data. You start with mathematical models (geometry, lighting physics) and output an image. Example: A video game engine rendering a 3D dragon model onto your 2D screen.
- ❖ **Image Processing:** This takes an image as input and outputs a modified image. The goal is to enhance or change the visual quality. Example: Adjusting the brightness of a photo, removing "red-eye," or blurring a background.
- ❖ **Computer Vision (CV):** This takes an image as input and outputs information or understanding. The goal is for the computer to "see" and interpret the content. Example: Feeding a photo of a street to a system, and the system outputting the text: "There are 3 cars and 1 pedestrian."

Computer Vision Core Concepts

❖ Image Filtering and Enhancement

- Before a computer can analyze an image, it often needs to "clean" it.
- Filtering involves modifying pixels based on their neighbors.
- For example, a Gaussian Blur filter might be used to reduce "noise" (graininess) in a low-light photo so that the important shapes are clearer. Another common technique is Histogram Equalization, which boosts the contrast of an image so that details in shadows become visible.

❖ Feature Extraction (Edges)

- Once the image is clean, the computer looks for "features"—interesting parts of the image that act as landmarks.
- The most fundamental feature is an Edge.
- Algorithms like the Canny Edge Detector look for sharp changes in brightness (e.g., where a white shirt meets a black background) to outline objects.
- More complex feature extractors, like SIFT (Scale-Invariant Feature Transform), can identify corners and blobs that stay recognizable even if the image is rotated or zoomed in.

Computer Vision Core Concepts

❖ Object Detection and Recognition

- This is the final high-level step.
- Object Detection asks "Where is it?" and draws a bounding box around objects (e.g., drawing a box around every face in a crowd).
- Object Recognition asks "What is it?" and assigns a label (e.g., identifying that the face belongs to "John Doe"). Modern systems use Convolutional Neural Networks (CNNs) like YOLO (You Only Look Once) to do both simultaneously in milliseconds.

Applications of Robotics

- ❖ **Industrial Automation:** This is the largest sector. Robots handle "3D" jobs: Dull, Dirty, and Dangerous. For example, in an automotive plant, robotic arms perform spot welding with perfect precision 24/7, or paint car bodies without exposing humans to toxic fumes.
- ❖ **Surgical Assistance:** Robots like the da Vinci Surgical System allow surgeons to operate with superhuman precision. The surgeon controls the robot from a console; the robot's tiny hands can make incisions smaller than a human hand ever could, leading to faster patient recovery.
- ❖ **Space Exploration:** Robots go where humans cannot survive. The Mars Rovers (like Perseverance) are autonomous robots that navigate the Martian surface, drill into rocks, and analyze soil samples to send data back to Earth, surviving extreme radiation and temperature changes.

Applications of Computer Vision

- ❖ **Autonomous Vehicles:** Self-driving cars (like those from Waymo or Tesla) rely heavily on CV. They use cameras to detect lane lines to steer, read traffic signs to know the speed limit, and identify pedestrians to brake automatically.
- ❖ **Facial Recognition:** This is used for security and convenience. Phones use it to unlock your screen by mapping the 3D geometry of your face. Airports use it at "e-gates" to verify that your face matches your passport photo without a human officer checking.
- ❖ **Medical Imaging Analysis:** CV helps radiologists by acting as a "second pair of eyes." AI algorithms scan X-rays and MRIs to highlight potential tumors or fractures that are too faint for the human eye to catch easily, increasing diagnostic accuracy.
- ❖ **Quality Control:** In factories, high-speed cameras watch assembly lines. CV systems instantly analyze every product passing by (e.g., a bottle on a conveyor belt) to detect defects like a missing cap or a torn label, triggering an actuator to kick the bad product off the line

Sustainable AI

- ❖ Sustainable AI refers to designing, training, deploying, and operating AI systems in ways that minimize environmental impact (energy, carbon, water, e-waste) while maximizing societal and economic benefit — essentially applying sustainability principles across the full AI lifecycle.
- ❖ Two foundational paradigms:
 - **AI for sustainability** — using AI as a tool to solve environmental/resource problems (grids, climate, agriculture, conservation).
 - **Sustainability of AI** — reducing the environmental footprint AI itself creates (compute, energy, hardware, water for cooling).



Green-by AI: Climate Action & Optimization

- ❖ AI serves as a powerful optimization tool to accelerate decarbonization and improve resource distribution.
- ❖ **Energy Grid Optimization & Smart Grids**
 - LSTM/Transformer-based load forecasting to balance supply and demand in real time.
 - ML models predicting solar/wind output variability to smooth intermittency (e.g., short-term irradiance/wind forecasting).
 - Anomaly detection models flagging transformer failures or transmission losses before they cascade.
- ❖ **Climate Change Prediction**
 - AI (CNNs, GANs) used to convert coarse global climate model outputs into high-resolution regional predictions.
 - Deep learning for hurricane path prediction, flood risk mapping
 - ML for tracking and predicting carbon flux from satellite + sensor data.
- ❖ **Resource Efficiency**
 - Computer vision + sensor fusion for irrigation optimization, reducing water waste.
 - AI reducing material waste in manufacturing via predictive maintenance and process tuning.
 - Predictive models for leak detection in water networks; AI-driven waste sorting/recycling (robotic sorting with CV).

GREEN AI

- ❖ Prioritizing efficiency alongside performance metrics
 - Quantization & Pruning
 - Distillation & Sparse MoE
 - Carbon-aware scheduling
- ❖ Carbon Footprint: Training vs. Inference
 - "Green AI" advocates treating computational cost, energy consumption, and carbon impact as primary evaluation metrics alongside accuracy.
 - Training a multi-billion parameter Large Language Model (LLM) consumes hundreds of megawatt-hours of electricity.
- ❖ Environmental Cost Factors
 - Total carbon emissions E_{CO_2} depend on operational energy E and the local grid's carbon intensity factor I_{grid}
 - Data centers use evaporative cooling towers and thermal power plants that consume billions of gallons of fresh water annually to keep servers from overheating.
- ❖ Technical Solutions for Green AI
 - Quantization: Reducing weight precision from FP32 (32-bit floating point) to INT8 or INT4, reducing memory bandwidth and energy usage per operation.
 - Pruning: Removing redundant weights or neural connections without losing overall performance.
 - Knowledge Distillation: Training a smaller, lightweight "student" model to replicate the behavior of a massive "teacher" model.
 - Advanced Cooling Systems: Transitioning data centers to closed-loop liquid cooling or immersion cooling to drastically reduce freshwater consumption.